Interval certification of selected equal-mass three-body periodic brake orbits

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Abstract

We give interval-arithmetic existence certifications for parameter boxes associated with 13 of the first 20 rows of a published table of equal-mass planar three-body periodic brake orbits. An Arb implementation certifies rows 1–6 and 10–16. Separate flow computations certify ten of these rows: a handwritten MPFR integrator certifies rows 1, 2, 3 and 12, and a CAPD multiprecision C1-Lohner implementation certifies rows 4, 5, 6, 10, 13 and 14. Its six selected paths have source and saved-trace audits of the local inclusions and their connection to the original family. Equal centers, nested parameter boxes and uniqueness identify the same normalized solution in each pair. The argument combines quarter-period symmetry, validated flow and derivative enclosures, a three-variable Krawczyk contraction and positive pairwise-distance bounds. The CAPD route uses nonuniform time steps and forward elimination of the multiple-shooting connection derivatives. Each certificate gives a unique reduced zero in its stated box and a nonconstant, collision-free periodic solution. The implementations share the model and symmetry proof. Row counts do not establish geometric distinctness; least period, stability and catalogue completeness are outside the claim. Seven entries numerically match previously catalogued self-dual candidates; earlier labels for rows 4, 10 and 12 were not identified in the tables reviewed. We make no claim of priority.

1 The question and the result

High-precision numerical data locate a candidate periodic orbit, but a small return residual alone does not establish the existence of an exact periodic solution. We study this question for the Euler-coordinate data of Hristov et al. [1, 2]. The concrete object to be certified is a parameter box around the finite decimals in a table row. The numerical discovery of these candidates belongs to the catalogue authors. A zero of a reduced quarter-period equation supplies the exact parameters; a symmetry argument then supplies the full periodic solution.

Theorem 1 (Computer-assisted result)**.** *Let $x_{0,j} = (a_{0,j}, b_{0,j}, T_{0,j})$ be the exact rational triple given by the first three decimal fields of row j of* `euler_100_4860.txt`*, with row numbers starting at one. For*

$$\mathcal{J} = \{1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 15, 16\},$$

the Arb computations certify exactly one zero $x_j = (a_j, b_j, T_j)$ of the map F defined in Section 2 in

$$X_j = x_{0,j} + [-r_j, r_j]^3, \qquad r_j = \begin{cases} 2^{-120}, & j = 2, 3, \\ 2^{-160}, & j \in \mathcal{J} \setminus \{2, 3\}. \end{cases}$$

The solution with these parameters is nonconstant, is collision-free for all real time, and has $T_j > 0$ as a period. Its pairwise distances have the positive lower bounds in Table 4.

For $j \in \{1, 2, 3, 4, 5, 6, 10, 12, 13, 14\}$, a second flow implementation also certifies exactly one zero in $Y_j = x_{0,j} + [-s_j, s_j]^3$, where

$$s_j = \begin{cases} 2^{-90}, & j = 4, 13, \\ 2^{-100}, & j = 1, 2, 3, 12, 14, \\ 2^{-130}, & j = 6, \\ 2^{-140}, & j = 5, 10. \end{cases}$$

The second computation uses the handwritten MPFR integrator for $j = 1, 2, 3, 12$ *and CAPD for* $j = 4, 5, 6, 10, 13, 14$. *In each case its zero is* x_j.

The uniqueness assertion concerns F with labels, Euler initial configuration, orientation and scale fixed. It is not uniqueness in the space of all periodic three-body motions. A certified T_j need not be the least period. Thus the theorem concerns 13 row-associated boxes, without counting geometrically distinct orbits. The numerical part of the proof is the set of validated flow and derivative enclosures. The analytic implications of those enclosures are proved below.

1.1 Scope of assurance

Table 1 gives the current guarantee structure. The three implementations use a common model, reduced map and prose symmetry proof. Arb and handwritten MPFR share the Taylor sufficient conditions below; CAPD supplies a different flow-enclosure route. Code separation therefore tests implementation agreement under common mathematical premises. The MPFR and CAPD routes both use MPFR scalar arithmetic. This is a computer-assisted prose argument with the trusted numerical components stated in the table.

Component	Present assurance and extent
Quarter-period reduction	Prose proof in Lemma 2; the lemma is a common premise of all three implementations.
Flow and derivative	Arb validates the 13 stated rows; MPFR separately validates four; CAPD validates six further rows. Outward arithmetic, the respective integrator, compiler and runtime belong to the trusted computing base.
Terminal algebra	Exact rational checks of the saved intervals verify the finite matrix and contraction tests. They take the saved flow and derivative enclosures as inputs.
Agreement	Equal rational centers, nested boxes and uniqueness identify the same reduced zero for rows 1, 2, 3, 4, 5, 6, 10, 12, 13 and 14.
CAPD audit	Separate integer/rational checks of the saved local inclusions, time choices, derivative reduction and terminal data cover the six selected paths. Taylor coefficients and set updates retain a fixed-source trust premise; these audits are not new ODE integrations.
Formal and human review	No Lean formalization and no human expert review of this work. The scope and outcome of AI mathematical reading are recorded in the accompanying verification record.
Claim boundary	Existence, a positive collision margin and local parameter-box uniqueness; least period, stability, geometric distinctness and completeness of the table remain open here.

Table 1: The layers of the computer-assisted argument.

The audits do not independently regenerate Taylor coefficients, rebuild the cloud binaries, or formally verify the software. Compiler, MPFR/GMP, CAPD arithmetic, automatic differentiation and set operations, hardware, and the correspondence between fixed sources and recorded executions remain trusted. The methods share modeling and symmetry premises; prior exposure to some results prevents a claim of complete blinding. For rows 4 and 14, the terminal matrix J comes from the saved derivative reduction, not the final global matrix. Only the row-10 direct path is used here. The general step-shrinking rounding counterexample remains valid; the six selected paths use audited normal accepted calls. Rows 11, 15 and 16 retain Arb-only certification; the second-route certificates for rows 11 and 16 remain on hold.

Exact artifact locations, input and source hashes, settings, retained unsuccessful attempts, comparison records, and the provenance of the mathematical reading are documented in the accompanying `VERIFICATION-RECORD.md`.

2 Model and quarter-period symmetry

All three masses and the gravitational constant are one. Positions and velocities are $q_i, p_i \in \mathbb{R}^2$, with

$$\dot{q}_i = p_i, \qquad \dot{p}_i = \sum_{k \neq i} \frac{q_k - q_i}{\|q_k - q_i\|_2^3}. \tag{1}$$

For $x = (a, b, T)$ we impose

$$\begin{aligned} &q_1(0) = (-1, 0), \quad q_2(0) = (1, 0), \quad q_3(0) = (0, 0), \\ &p_1(0) = p_2(0) = (a, b), \qquad p_3(0) = -2(a, b). \end{aligned} \tag{2}$$

The reduced map is

$$F(a, b, T) = \big(p_{1x}, p_{1y}, p_{2x}\big)(T/4). \tag{3}$$

It is defined on any parameter box for which the required flows exist without collision through all relevant terminal times.

Each input row has fields (a_0, b_0, T_0, T^*). We integrate through a quarter of the third field T, rather than the fourth, scale-invariant period T^*. The decimal strings are exact rational centers, not assumed error bounds on the unknown exact solution.

Lemma 2 (Three terminal components suffice)**.** *Suppose a solution of* (1)–(2) *is collision-free on the closed interval* $[0, \tau]$*, with* $\tau > 0$*. If*

$$p_{1x}(\tau) = p_{1y}(\tau) = p_{2x}(\tau) = 0, \qquad q_{2x}(\tau) - q_{3x}(\tau) \neq 0, \tag{4}$$

then it extends to a collision-free solution on $\mathbb{R}$ *with period* 4τ*. A common lower bound* $d > 0$ *for all three pairwise distances on* $[0, \tau]$ *holds on the full extension. If* $(a, b) \neq (0, 0)$*, the solution is nonconstant.*

Proof. Linear momentum $P = \sum_i p_i$ and scalar angular momentum $L = \sum_i q_i \times p_i$ vanish initially. Pairwise cancellation of forces gives $\dot{P} = 0$, and the central-force identity gives $\dot{L} = 0$. The center of mass remains zero, so $q_3 = -q_1 - q_2$ for the whole interval. At τ, the three zero components in (4) and $P = 0$ imply $p_{3x} = 0$ and $p_{3y} = -p_{2y}$. Thus $p_{ix}(\tau) = 0$ for all i and $p_{1y}(\tau) = 0$, so all $q_{iy}p_{ix}$ terms in the cross products vanish. Hence

$$0 = L = (q_{2x} - q_{3x})p_{2y}.$$

The nonzero factor in (4) forces $p_{2y} = p_{3y} = 0$. Thus all velocities vanish at τ.

Time reversal about this brake instant defines $q(\tau + s) = q(\tau - s)$ for $0 \leq s \leq \tau$. It joins with zero velocity and solves the same second-order equation. Uniqueness on the collision-free domain makes it the continuation of the original solution. Its state at 2τ is $(q(0), -p(0))$.

Define the involution on triples of planar vectors by

$$S(z_1, z_2, z_3) = (-z_2, -z_1, -z_3).$$

It is the composition of a swap of the first two equal masses and a rotation through 180 degrees, and is a symmetry of the equations. For (2), $Sq(0) = q(0)$ and $Sp(0) = -p(0)$. Define the second half of the solution by

$$(q(2\tau + s), p(2\tau + s)) = S(q(s), p(s)), \qquad 0 \leq s \leq 2\tau.$$

The initial state of this half equals the state already reached at 2τ, so uniqueness again gives a smooth continuation. At 4τ the state is $S(q(0), -p(0)) = (q(0), p(0))$. Periodic continuation defines the solution for every real time. Time reversal and S preserve the set of pairwise distances, so the common lower bound persists. A nonzero initial velocity proves nonconstancy. □

The nondegeneracy test is essential: three zero velocity components without the nonzero factor in (4) would not force all velocities to vanish. In center-of-mass coordinates the tested quantity is $q_{2x} - q_{3x} = q_{1x} + 2q_{2x}$. Under the hypotheses of Lemma 2, a zero of F determines a collision-free periodic solution.

3 Validated integration and the finite-dimensional test

We describe sufficient conditions used by the computations. Every interval operation in this section is an inclusion operation. Matrix norms are induced infinity norms, and boxes are closed rectangular interval sets. A positive distance bound is checked on a whole time slab, not on sampled trajectory points.

3.1 An eight-dimensional center-of-mass system

Write

$$y = (x_1, y_1, x_2, y_2, u_1, w_1, u_2, w_2), \qquad q_3 = -q_1 - q_2, \quad p_3 = -p_1 - p_2.$$

With $d_{12} = q_2 - q_1$, $d_{13} = -2q_1 - q_2$ and $d_{23} = -q_1 - 2q_2$, set $g(d) = d/(d \cdot d)^{3/2}$. Then

$$\dot{p}_1 = g(d_{12}) + g(d_{13}), \qquad \dot{p}_2 = -g(d_{12}) + g(d_{23}).$$

Together with $\dot{q}_i = p_i$, these equations define $y' = f(y)$ and recover (1). For

$$B(d) = (d \cdot d)^{-3/2} I_2 - 3(d \cdot d)^{-5/2} d d^{\mathsf{T}},$$

the Jacobian has block form

$$A = Df = \begin{pmatrix} 0 & I_4 \\ H & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} -B_{12} - 2B_{13} & B_{12} - B_{13} \\ B_{12} - B_{23} & -B_{12} - 2B_{23} \end{pmatrix}, \tag{5}$$

where $B_{ij} = B(d_{ij})$. Every evaluation requires a strictly positive lower bound for each $d_{ij} \cdot d_{ij}$. Polynomial squaring and positive square roots are evaluated with their respective inclusion rules.

3.2 Existence and variational bounds on one slab

Let Y enclose all starting states of a step of rational length $h > 0$. Choose a collision-free box B and a fixed positive diagonal matrix D for which

$$Y + [0, h] f(B) \subseteq B, \qquad \alpha \geq \sup_{z \in B} \left\| D^{-1} A(z) D \right\|_\infty, \qquad h\alpha < \tfrac{1}{4}. \tag{6}$$

For any starting state in Y, the Picard operator maps continuous B-valued paths into themselves. In the weighted sup norm $\sup_t \left\| D^{-1} y(t) \right\|_\infty$ it is a contraction with constant at most $h\alpha$. Thus a unique solution exists on the entire slab and stays in B.

Let M be the derivative of the step flow, with $M(0) = I_8$. Its equation is $M' = A(y)M$. The integral equation, in the same weighted norm, gives

$$\sup_{0 \leq t \leq h} \left\| D^{-1} M(t) D \right\|_\infty \leq \frac{1}{1 - h\alpha}, \qquad \left\| D^{-1} (M(t) - I_8) D \right\|_\infty \leq E := \frac{h\alpha}{1 - h\alpha}. \tag{7}$$

In particular, a componentwise enclosure for every such $M(t)$ is

$$\mathcal{M}_{ij} = \delta_{ij} + [-E D_{ii}/D_{jj}, E D_{ii}/D_{jj}]. \tag{8}$$

The unweighted choice is $D = I_8$. The weighted implementation uses $D = \operatorname{diag}(I_4, w I_4)$ with a positive point dyadic w fixed throughout each slab. By (5), a certified bound $R_H \geq \sup_B \|H\|_\infty$ supplies

$$\alpha = \max(w, R_H / w). \tag{9}$$

When $R_H > 0$, one may choose a dyadic $w \geq \sqrt{R_H}$; for $R_H = 0$, one may take $w = 1$. The method also permits $w < 1$. This changes the bound used for a local proof, not the vector field. Since D is fixed within the slab, there is no derivative of D in the variational equation. Taylor remainders and terminal contractions are tested in the original coordinates.

3.3 Taylor remainders and propagation

Normalized Taylor coefficients c_k of the state satisfy

$$(k+1)c_{k+1} = [s^k] f\left(\sum_{j=0}^{k} c_j s^j\right). \tag{10}$$

If $A(y(s)) = \sum A_k s^k$, the variational coefficients satisfy $(k+1)M_{k+1} = \sum_{j=0}^{k} A_j M_{k-j}$. The recurrences are evaluated with outward interval operations.

For an order N step from a point c, evaluate the state polynomial through degree $N-1$ at c and enclose its degree N remainder using the coefficient evaluated on B. Thus an enclosure C of $\phi_h(c)$ is

$$C = \sum_{k=0}^{N-1} c_k(c)h^k + c_N(B)h^N. \tag{11}$$

This follows componentwise from Taylor's theorem, since all intermediate states belong to B. Apply the same argument to the joint system $(y, M)' = (f(y), A(y)M)$. Its initial set is (Y, I_8) and its remainder set is $(B, \mathcal{M})$. This gives an interval matrix $\mathcal{D}$ containing $D\phi_h(z)$ for every $z \in Y$. The local remainder tolerance is an additional acceptance criterion; the existence and inclusion arguments are (6)–(11).

To control wrapping, represent the starting set by $c+Qb$, where Q is an invertible point matrix and b is a zero-centered box; choose Y as a rectangular enclosure. Track the center-parameter solution separately in $c + Qp$, maintaining $p \subseteq b$. The mean-value formula on the convex box Y gives

$$\phi_h(c + Qb) \subseteq C + \mathcal{D}Qb.$$

Choose a new point center $c_+ \in C$ and point basis Q_+. An interval inverse of Q_+ yields zero-centered enclosures

$$\begin{aligned} b_+ &\supseteq Q_+^{-1}(C - c_+ + \mathcal{D}Qb), \\ p_+ &\supseteq Q_+^{-1}(C - c_+ + \mathcal{D}Qp). \end{aligned} \tag{12}$$

A floating-point QR factorization may propose a basis, but its orthogonality is not an assumption: invertibility and the inverse enclosure are checked for the chosen point matrix.

The 8×2 sensitivity $W = \partial y/\partial(a, b)$ starts with four zero position rows and velocity rows $(1,0),(0,1),(1,0),(0,1)$. If $W \in QR_W$, update $R_{W,+} \supseteq Q_+^{-1}\mathcal{D}QR_W$. This encloses the derivative for all parameters in the box, not only its center.

3.4 The variable terminal time

The integrator reaches an exact dyadic time t_0 near $T_0/4$. Every step length is rational and positive, and the saved lengths sum exactly to t_0. For $X = x_0 + [-r, r]^3$, choose

$$\Delta \geq |T_0/4 - t_0| + r/4. \tag{13}$$

An endpoint box B_e is accepted when

$$Y_e + [-\Delta, \Delta] f(B_e) \subseteq B_e, \qquad \Delta \sup_{B_e} \left\| D^{-1} A D \right\|_\infty < \tfrac{1}{4}, \tag{14}$$

where Y_e encloses the parameter-box states at t_0. The positive- and negative-time Picard arguments then cover all required terminal times. If δ_0 encloses the exact rational $T_0/4 - t_0$, the center solution at $T_0/4$ belongs to $c + Qp + \delta_0 f(B_e)$. Its three selected velocities give an enclosure F_0 of $F(x_0)$. Errors from converting the decimal centers to binary are included in p and δ_0.

For completeness, let W_0 enclose $W(t_0)$, choose $\alpha_e \geq \sup_{B_e} \left\| D^{-1} A D \right\|_\infty$, and put $E_e = \Delta\alpha_e/(1-\Delta\alpha_e)$. If $n \geq \left\| D^{-1} W_0 \right\|_\infty$, an enclosure throughout the small terminal interval is

$$W_e = W_0 + \{V : |V_{ij}| \leq D_{ii} E_e n\}.$$

An endpoint derivative enclosure is consequently $W_0 + [-\Delta, \Delta] A(B_e) W_e$. Selecting its velocity rows $(1x, 1y, 2x)$ gives the a, b columns of an interval matrix J. The T column consists of the same components of $f(B_e)/4$. The chain rule therefore gives $DF(X) \subseteq J$.

3.5 Krawczyk contraction and collision margins

Choose an invertible point matrix $C_0 \in \mathbb{R}^{3\times 3}$, put $Z = I_3 - C_0 J$, and form the centered Krawczyk enclosure

$$K_c = -C_0 F_0 + Z[-r, r]^3. \tag{15}$$

The finite-dimensional tests are

$$K_c \subset \operatorname{int}[-r, r]^3, \qquad \|Z\|_\infty < 1. \tag{16}$$

Proposition 3. *Suppose F is continuously differentiable on a neighborhood of X, $F(x_0) \in F_0$, $DF(X) \subseteq J$, C_0 is invertible, and* (16) *holds. Then F has exactly one zero in X.*

Proof. For $G(x) = x - C_0 F(x)$, the integral mean-value formula shows $G(X) \subseteq x_0 + K_c \subseteq \operatorname{int} X$. Also $DG(X) \subseteq Z$, so G is a contraction on the complete space X. Banach's theorem gives exactly one fixed point in X. Invertibility of C_0 identifies fixed points of G with zeros of F. □

The validated flow on collision-free slabs provides the smoothness needed here.

Every slab and the terminal box must have a common positive rational lower bound d_2 for all three squared pairwise distances. A displayed distance bound d is rounded down so that $d^2 \le d_2$. We also require that the interval for $q_{1x} + 2q_{2x}$ on B_e exclude zero, that the parameter box have $T > 0$, and that its initial velocity exclude $(0, 0)$. At the zero supplied by Proposition 3, Lemma 2 then gives the claimed nonconstant periodic solution and the full-time collision margin.

4 Separate implementations and identification of the zeros

Implementation A uses Python 3.14.7, python-flint 0.9.0 and FLINT 3.6.0 with Arb real balls. Implementation B is a separate C++ program using MPFR 4.2.2 and GMP 6.3.0 with lower and upper interval endpoints. Both implement the Taylor method of Section 3, with separate interval and basis handling. Implementation C uses CAPD 6.1.0 with multiprecision enabled, MPFR 4.2.0 and GMP 6.2.1. Its C1-Lohner integration, normalized clock and derivative reduction are described below. A versus B and A versus C provide two flow routes per doubly certified row, using three codebases overall.

Run family	Bits	Order	Parameter radius	Tolerance
A, rows 2, 3	320	24	2^{-120}	2^{-170}
A, other successes	448	48	2^{-160}	2^{-250}
B, all successes	384	28	2^{-100}	2^{-180}
C, rows 4, 13, 14: C^0	320	24	Point enclosure	2^{-220}
C, rows 4, 13, 14: C^1	256	24	2^{-90} or 2^{-100}	2^{-180}
C, rows 5, 6, 10: C^0	448	56	Point enclosure	2^{-300}
C, rows 5, 6, 10: C^1	320	48	2^{-130} or 2^{-140}	2^{-260}

Table 2: Selected successful settings. A uses an unweighted Picard norm on rows 2 and 3 and a fixed diagonal weight on the other rows; B uses the unweighted norm. C uses CAPD doubleton enclosures. These run families are not a common-setting benchmark.

4.1 CAPD: a fixed clock for a variable period

Adjoin the period as a constant ninth state, $\theta = T$, and solve

$$\frac{d}{ds}(y, \theta) = \big((\theta/512) f(y), 0\big), \qquad 0 \le s \le 128. \tag{17}$$

For each initial parameter triple the physical time is $t = \theta s/512$, so the fixed endpoint $s = 128$ is exactly $t = T/4$. The entries p_i remain physical velocities. The affine initial map $E(a, b, T)$ encodes (2) and $\theta = T$; a three-column doubleton carries the correlated parameter box $E(Y_j)$. Outward conversion of the

exact rational center is included. The center computation encloses $E(x_{0,j})$ and supplies $F(x_{0,j})$; the box computation supplies the derivative over every parameter in Y_j.

The nonvalidated reference trajectory serves progress and cost estimation. CAPD's step controller selects the accepted steps; reference samples neither prescribe a certified partition nor replace the evolving initial family. In particular, a reference interval count must not be identified with the number of accepted slabs.

Let H_k be the saved positive interval of step lengths. The audits establish that every nonfinal H_k is a singleton. Choose its rational value h_k, and choose the final length as

$$h_{m-1} = 128 - \sum_{k<m-1} h_k.$$

The final choice is positive and belongs to the final H_{m-1}; it need not equal its left endpoint. Put $s_0 = 0$ and $s_{k+1} = s_k + h_k$. These are exact rational times common to all parameters, and $s_m = 128$. Membership of a length in H_k is used only after a uniform local inclusion over that interval has been established.

For the selected normal calls, let Π_k enclose the Taylor polynomial on $[0, \sup H_k]$, M_k be its remainder margin, $\mathcal{E}_k$ the candidate enclosure, and R_k the remainder evaluated on that enclosure. The state and local variational components satisfy

$$R_k \subset \operatorname{int} M_k, \qquad \Pi_k + M_k \subseteq \mathcal{E}_k, \qquad 0 \in \operatorname{int} M_k.$$

The time-zero input state box and variational identity matrix are contained in the corresponding polynomial ranges and enclosures. Strictly positive squared distances on the state tube make the extended ODE smooth there. Its Taylor formula and a first-exit argument give enclosure on $[0, \sup H_k]$, uniformly for every $h \in H_k$. The local 9×9 matrix L_k encloses the flow derivative for the input box and every such h, including the variational remainder. The fixed-source audit connects these polynomial and remainder records to CAPD's Taylor/AD calculation; the separate integer checks verify their inclusions.

To connect these local statements to the original parameters, write $u = (a, b, T) \in Y_j$, $S = DE$, and, for the normalized flow Φ,

$$Y_k(u) = \Phi_{s_k}(E(u)), \qquad V_k(u) = D_u Y_k(u).$$

At each selected time, both the cached axis box B_k and the geometric doubleton $\mathcal{R}_k$ contain $Y_k(u)$. They need not contain each other. The set update preserves this reachable-family assertion; no assertion that the saved output contains the image of every fictitious point of B_k is required. The matrix family carried by the set update contains a true matrix $A_k(u)$ with

$$A_k(u)S = V_k(u), \qquad A_0 = I_9.$$

The actual derivative obeys the chain rule at the same u, s_k, h_k. Thus the local product and the saved global 9×3 enclosure G_{k+1} both contain $V_{k+1}(u)$. This simultaneous state and tangent induction, starting with the original affine family, justifies their intersection. It does not reset a state or a derivative at a reference sample. The center calculation has the same time-coverage construction but starts with the center enclosure only, independently of the command line's radius label.

4.2 Eliminating the connection variables

Let $\phi_k(z_k, T) - z_{k+1} = 0$ be the eight-state connection equation between consecutive nodes. With the initial node prescribed by the initial map, the Jacobian with respect to the subsequent nodes $z_1, \ldots, z_m$ is block lower triangular, with diagonal blocks $-I$. Forward elimination of these variables transports the derivative with respect to (a, b, T), retaining the parameter dependence of the prescribed initial node. Using the ninth constant state includes the direct period dependence:

$$D_0 \supseteq DE, \qquad D_{k+1} = \operatorname{out}(L_k D_k) \cap G_{k+1}. \tag{18}$$

Here D_k is an interval enclosure of $V_k(u)$ for every original parameter u. Interval products are rounded outward onto the 2^{-240} grid for rows 4, 13 and 14, and the 2^{-512} grid for rows 5, 6 and 10. Intersecting each propagated enclosure with CAPD's global parameter-derivative enclosure preserves inclusion, because both contain the true derivative for the same initial family and time node. The final rows (p_{1x}, p_{1y}, p_{2x}) give a 3×3 enclosure J of $DF(Y_j)$.

The proof uses coverage of every true trajectory starting in $E(Y_j)$. Consequently the reduced derivative is a derivative enclosure on the whole original parameter box. Uniqueness of an isolated root of an expanded shooting system alone would not give that coverage. The reduction uses local and global enclosures from the same CAPD flow; they are not two independent integrations.

The terminal MPFR calculation uses 512 bits for rows 4, 13 and 14, and 1024 bits for rows 5, 6 and 10, and applies (15)–(16). The audits separately propagate a reduced enclosure using a 2^{-512} grid and verify that the saved terminal derivative contains it. For rows 4, 13 and 14 this is an inclusion, not numerical equality of the old and audited reductions. The terminal J is linked to the velocity rows of the saved reduction; the terminal F_0 and nondegeneracy factor are linked to the center and box endpoints, respectively. Exact rational checks of these saved terminal inputs establish invertibility, strict inclusion, contraction and the nonzero factor. The six local-inclusion audits supply the additional connection to the flow described above; terminal algebra alone would not supply it.

4.3 Identifying the two zeros for each row

Proposition 4 (Identification by nested boxes)**.** *Suppose the same reduced equation F has one certified zero in each of two boxes $X \subseteq Y$, and has exactly one zero in Y. Then the two zeros coincide.*

Proof. The zero in X belongs to Y and must equal its unique zero. □

For all ten rows, the saved comparisons establish equal exact rational centers. The radii in Theorem 1 give $X_j \subseteq Y_j$. Proposition 4 therefore identifies the two zeros. A and B cover the variable endpoint by (13); C covers it by (17). All represent the same map F. Overlap of endpoint or residual intervals is a useful consistency check, but nesting and uniqueness are the reason for the same-solution conclusion.

Rows 1, 2, 3 and 12 have B's unweighted Taylor route. Rows 4, 5, 6, 10, 13 and 14 have C's CAPD route. These ten second-route certificates do not rely on A's weighted integration bounds. Rows 11, 15 and 16 remain certified by A alone in the adopted evidence; partial CAPD records on rows 11 and 16 do not add a second certificate.

5 Results by table row

The present result set concerns rows 1–20 of the 4,860-row input. Table 3 distinguishes certification, unsuccessful terminal tests, partial integration and unstarted runs. Counts describe coverage over retained attempts with varying settings, rather than a success rate under a single fixed protocol.

Row	A (Arb)	B (MPFR)	C (CAPD)	Two routes, same zero
1	C	C	N	Yes
2	C	C	N	Yes
3	C	C	N	Yes
4	C	P	C	Yes
5	C	P	C	Yes
6	C	P	C	Yes
7	U	P	P	—
8	U	P	N*	—
9	U	P	N*	—
10	C	P	C	Yes
11	C	P	H	—
12	C	C	N	Yes
13	C	P	C	Yes
14	C	N	C	Yes
15	C	N	U/P	—
16	C	N	H	—
17	U	N	P	—
18	U	N	N*	—
19	U	N	N*	—

Row	A (Arb)	B (MPFR)	C (CAPD)	Two routes, same zero
20	U	N	P	—
Certified rows	13	4	6	10

Table 3: Status over retained attempts. C: certified; U: attempted but uncertified; P: incomplete flow; N: no full-flow attempt. N*: local pilot tests only. H: second certificate on hold, with incomplete audit evidence. For C on row 15, U/P retains terminal failures and incomplete integrations; it does not imply a second certificate. A dash makes no same-solution assertion.

For the distance columns, Table 4 gives downward-rounded linear bounds d_A, Table 5 displays approximations of $\sqrt{d_{2,B}}$, and Table 6 displays approximations of the squared bounds $d_{2,C}$.

Row	A steps	A radius	Full-time distance lower bound d_A
1	1,885	2^{-160}	0.287841
2	39,860	2^{-120}	0.014202
3	47,639	2^{-120}	0.015223
4	8,047	2^{-160}	0.000363
5	10,809	2^{-160}	0.002044
6	10,922	2^{-160}	0.002188
10	12,963	2^{-160}	0.000286
11	13,080	2^{-160}	0.000266
12	5,670	2^{-160}	0.287755
13	8,813	2^{-160}	0.002304
14	8,902	2^{-160}	0.001999
15	14,991	2^{-160}	0.000007
16	13,321	2^{-160}	0.000012

Table 4: Selected A certificates. Every decimal in the last column is a downward-rounded bound with $d_A^2 \le d_2$. The slab bound holds through the quarter-period and transfers to all time by Lemma 2. Different successful certificates for the same row can give different valid lower bounds.

Row	B steps	B radius	Approximate $\sqrt{d_{2,B}}$	Approximate contraction bound
1	2,326	2^{-100}	0.288723577	1.48×10^{-18}
2	31,139	2^{-100}	0.014202810	3.58×10^{-3}
3	38,951	2^{-100}	0.015223399	6.31×10^{-3}
12	6,995	2^{-100}	0.288635565	1.70×10^{-6}

Table 5: B certificates on four of the ten doubly certified solutions. The last two columns are display approximations of saved quantities; they are not rounded bounds used in the proof. Strict positivity, strict inclusion and contraction are decided on the saved binary intervals and their exact rational interpretation.

Row	C C^0 slabs	C C^1 slabs	Radius	Approximate $d_{2,C}$	q_C (approx.)
4	37,024	12,325	2^{-90}	$1.323016679 \times 10^{-7}$	0.1430438542
5	4,880	4,868	2^{-140}	$4.177760495 \times 10^{-6}$	0.0002962666
6	4,884	4,871	2^{-130}	$4.801121064 \times 10^{-6}$	0.1317961327
10	5,752	5,849	2^{-140}	$8.134207033 \times 10^{-8}$	0.0003566111
13	36,489	12,134	2^{-90}	$5.315061159 \times 10^{-6}$	0.0298201405
14	36,875	12,264	2^{-100}	$4.028534374 \times 10^{-6}$	0.0102606219

Table 6: C certificates. The squared-distance bounds and contraction bounds $q_C = \|I - C_0 J\|_\infty$ are display approximations; the contraction column transcribes the saved terminal field `q_upper`. This is distinct from the full Krawczyk inclusion ratio defined in Section 6. The saved exact intervals decide the inequalities. Strict Krawczyk inclusion and the nonzero endpoint factor also pass in all six cases. C^0 counts include reused center evaluations, which are independent of box radius.

A attempts every row from 1 to 20. Its seven uncertified rows are 7, 8, 9 and 17–20. Minimum-step stops and time limits give no nonexistence conclusion. A positive distance bound on a completed prefix does not bound an unintegrated portion of the quarter-period.

For B, rows 4–11 and 13 remain partial, and rows 14–20 have no full-flow attempt. Historical C tests on rows 5, 6, 10, 11 and 16 failed at radii 2^{-90} and 2^{-100}. The smaller-box successes for rows 5, 6 and 10 in Table 6 supersede those failures only for the stated boxes and selected paths.

For rows 11 and 16, the available second-route evidence does not establish all required flow and terminal connections. Neither partial result is adopted as a second certificate.

Later full box attempts on rows 7, 15, 17 and 20 stopped at normalized times approximately 50.1671, 72.2639, 41.0290 and 41.3307 of 128; their full center runs did not start. Rows 8, 9, 18 and 19 have local pilots only. The earlier row-15 2^{-90} terminal failure and 2^{-100} stop after 26,140 slabs at time 108.646304 are retained. None of these incomplete or unsuccessful tests proves nonexistence. In particular, rows 11, 15 and 16 still have A's existence certificates.

5.1 Measured work and recorded cost

The CAPD phase times reported below are observations from the earlier order-24 runs for rows 4, 13 and 14; they do not include the later certifications of rows 5, 6 and 10. The selected B runs take approximately 241, 1,963, 2,338 and 507 seconds for rows 1, 2, 3 and 12. For rows 1 and 12 these values are taken from the certificates' elapsed-time fields; for rows 2 and 3 they are rounded worker wall times. C's successful C^1 native runs take about 3,704, 3,093 and 2,831 seconds for rows 4, 13 and 14, respectively. Each requires a center evaluation as well: about 3,758, 3,735 and 3,732 seconds, with reuse across radii at the same exact center. C's phase times exclude compilation, record checks and transfer. Different boxes, precisions and machines preclude a controlled speed comparison between implementations.

The historical eight-row C batch covering rows 4, 5, 6, 10, 11, 14, 15 and 16 has 22 completed integrations, one retained partial run and three platform preemptions. It uses one physical core and 1 GiB per flow process, with at most eight concurrent cores. The largest recorded peak resident memory among the completed native integrations is 63,672 KiB. The batch's conservative recorded cost estimate is US$2.393, including unsuccessful attempts and a US$0.50 transfer/storage allowance. It is an estimate, not an observed invoice. The row-13 full-box attempt and smaller-box certification have separate recorded estimates of US$0.340 and US$0.225; the latter reuses the earlier center evaluation. These amounts exclude earlier development, local pilots, the later row-5/6/10 smaller-box runs and subsequent method audits. They measure the indicated historical batches, not the total cost of this article. Shared credit-ledger balances and audit expenditure are different accounting units and are not added to these figures. Local-pilot extrapolations are not measurements of full-orbit speed or cost.

The table contains 4,840 further rows, 21–4,860, outside A's attempted range and outside the present claim. Extending a run or improving an enclosure may change an uncertified status. No conclusion about the full database follows from this finite prefix.

Proof of Theorem 1. For A and B, the full sequence of accepted slabs satisfies the Picard self-map and contraction tests, positive distance tests, and state and variational Taylor remainder bounds. Verified basis changes preserve the enclosure inductively. The final time correction gives F_0 and J. For C, the audited uniform local inclusions and reachable-family/tangent induction cover the original initial family at common rational times; connection elimination encloses its parameter derivative. Thus each route supplies the inclusions required in Proposition 3. The saved terminal tests pass exact rational checks of invertibility, strict interior inclusion and contraction. The positive distance bound, nonzero terminal factor, positive period and nonzero initial velocity complete the hypotheses of Lemma 2. Thus every stated box contains its unique reduced zero and supplies the periodic solution and collision margin. For the ten doubly certified rows, the saved exact-center and box-inclusion comparisons, together with Proposition 4, identify the zeros. □

6 Certificate format and levels of checking

A and B certificates are JSON objects containing the input hash, row number, exact decimal center, radius exponent, precision, Taylor order, norm choice, remainder tolerance, complete rational step list,

and its exact total time. Each certificate also contains the final point center and basis, the point and parameter-box enclosures, the sensitivity enclosure, F_0, J, C_0, K_c, Z, B_e, the common squared distance bound, and the nondegeneracy interval. An interval ball is saved as

$$m2^a \ \pm \ n2^b, \qquad m, a, n, b \in \mathbb{Z}, \quad n \geq 0, \tag{19}$$

with integers encoded as strings where necessary. Matrix entries are stored row by row. This preserves exact binary endpoints independently of a decimal display. Point matrices have radius zero.

For C, the contraction bound q_C and the full inclusion ratio $\rho_C = \max_i \sup_{u \in (K_c)_i} |u|/r$ are separate quantities. The saved fields are `terminal.mpfr.q_upper` and `terminal.fraction.max_inclusion_ratio_float`, respectively. The latter is a display approximation; exact interval and rational tests decide strict inclusion. For example, row 13 has approximately $q_C = 0.02982014046783021$ and $\rho_C = 0.02982014046810953$. Their agreement at the shorter precision of Table 6 does not identify the two quantities: ρ_C includes the center residual term in (15).

The terminal certificate serializes a finite algebraic implication conditional on its flow and derivative enclosures. A success flag alone does not establish those analytic premises. Three additional kinds of evidence must be distinguished: source examination, saved local-inclusion and reduction checks, and a fresh validated integration replay. For C, the six selected paths have the first two kinds, including all final accepted-call records and the uniform-H argument in Section 4.1. They do not have independent regeneration of Taylor coefficients or new ODE reintegration from these audits. The trust boundary is stated once in Section 1.1.

Separate full-flow computations by B and C cover the ten specified rows. C's separately written saved-record checks and source audits connect its local inclusions to its terminal enclosures; a terminal rational checker by itself cannot do so.

An unsuccessful Picard, tolerance or terminal inclusion test is inconclusive. In particular, failure of strict Krawczyk inclusion does not prove that no zero exists. The documented rounding counterexample in CAPD's general step-shrinking formula is retained; the result here relies on the selected normal accepted calls, not on a general claim that every step-selection branch is correct.

7 Relation to earlier work and remaining questions

Hristov, Hristova, Puzynina, Sharipov and Tukhliev [1] supply the numerical catalogue and its quarter-period search formulation; their Section 4.2 imposes the vanishing of all six velocity components. Their published Euler-coordinate table [2] supplies every center in this work. Montgomery's treatment of brake orbits [4, Section 3] explains the role of time reversal, equal-mass permutations and Euler configurations. Lemma 2 states and proves the sufficient direction used here, with the additional nondegeneracy check that makes three velocity components sufficient.

Computer-assisted existence proofs in the three-body problem have substantial precedents. Kapela and Zgliczyński [5, Sections 2–3] combine rigorous flow and derivative integration with interval Newton and Krawczyk methods in their proof of the figure eight and its local uniqueness. Kapela and Simó [6] develop computer-assisted existence and KAM stability methods and apply them to the figure eight and selected rotating eights. We use no stability conclusion from that work for the brake orbits studied here. Johansson's Arb library [7] supplies rigorous ball arithmetic through the python-flint interface [8]; MPFR supplies directed, correctly rounded scalar operations for B and C [9]. These contributions provide the objects, symmetry ideas, existence-proof methods and arithmetic on which this work depends.

Earlier existence results for periodic brake orbits include the isosceles solutions of Moeckel, Montgomery and Venturelli [11, Theorem 2 and Section 5]. Those solutions encounter binary collisions and are distinct from the collision-free objects certified here. The numerical collision-free searches of Li and Liao [12] and Hristov et al. [13, 1] supply the catalogue context. CAPD's rigorous flow and variational algorithms [10] supply C's numerical foundation.

7.1 Correspondence with earlier numerical catalogues

Among the thirteen Arb-certified entries in Theorem 1, ten—1, 2, 3, 4, 5, 6, 10, 12, 13 and 14—have certificates from two implementations for the same reduced zero. These entries belong to the central-symmetry free-fall catalogue of Hristov et al. [1, 2]. Rows 1, 2, 3, 5, 6, 13 and 14 numerically match

self-dual entries 1, 2, 3, 8, 9, 27 and 28 of the earlier free-fall catalogue [13, 3], respectively. Entry 8 is also listed in Tables 2 and 3 of the earlier paper; entry 9 is identified in its accompanying online catalogue. These correspondences identify the published numerical candidates; they are not additional validated orbit-identification theorems. The thirteen Arb certifications and the ten same-solution comparisons retain the scope stated in Theorem 1.

2025 row	Earlier full-table row	Earlier self-dual label
1	3	self-dual #1
2	32	self-dual #2
3	35	self-dual #3
4	Not matched	Earlier label unconfirmed
5	126	self-dual #8
6	129	self-dual #9
10	Not matched	Earlier label unconfirmed
12	Not matched	Earlier label unconfirmed
13	807	self-dual #27
14	810	self-dual #28

Table 7: Numerical correspondence status for the ten rows with two flow implementations. The earlier table [3] contains 24,582 entries in `sol_80.txt` and 236 self-dual entries in `sd_80.txt`; all row numbers are one-based. "Not matched" means no match in these reviewed earlier tables, not a new numerical discovery: all ten entries are published in the 2025 catalogue [2].

The comparison uses the paired Euler-coordinate and free-fall tables published with [1, 2]. This pairing provides the change of representation; the present comparison does not validate it by a new flow integration. For each of the seven numerical matches in Table 7, the four free-fall fields (two initial position coordinates, period and scale-invariant period) agree to an absolute difference below 5×10^{-79}. Agreement of finite decimals identifies a numerical catalogue entry, but supplies no error bound on an older approximation and no proof that it equals a certified root. The rigorous same-root argument in Section 4.3 instead concerns two certified boxes for the same 2025 row.

For the original seven compared rows (1, 2, 3, 4, 12, 13 and 14), we did not identify earlier rigorous existence proofs in the sources reviewed through 23 October 2026. A supplement for rows 5, 6 and 10, reviewed through 25 October 2026, likewise did not identify earlier rigorous existence proofs for those specified entries. These reviews distinguish figure-eight and isosceles collision-brake results from the present candidates. Citation-index access was incomplete, and an identified citing review and a related full text were not obtained. These are limits of the reviewed evidence; no claim of priority is made.

Conclusion. The retained evidence gives thirteen Arb-certified parameter boxes and ten same-zero comparisons using a second flow route. The six adopted CAPD paths are supported by the reachable-family and tangent argument and their selected-trace audits. Rows 11, 15 and 16 retain Arb-only certification; the second-route certificates for rows 11 and 16 remain on hold. These are bounded certification results for published numerical candidates; no count of geometrically distinct orbits, least-period result, stability result, catalogue completeness or priority claim follows.

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Use of AI. AI systems assisted with the research planning, prose mathematics, implementation, record checking, mathematical reading, comparison and drafting. The author is responsible for the claims and presentation; AI assistance is not human expert review or formal verification.

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